Combinatorics 7

Siddharth K

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Problem 1

Each coefficient, n, in the generating function counts the number of sequences of length 14 whose entries are in $\{0, 3, 8\}$ and whose sum is n. The coefficient of x^{30} is

$$\binom{14}{4, 10, 0} + \binom{14}{9, 2, 3} = 21021.$$

Problem 2

Given that $a_{n+1} = 2a_n + 3^n$, multiply the recurrence by x^{n+1} and sum over n:

$$\sum_{n=0}^{\infty} a_{n+1}x^{n+1} = 2\sum_{n=0}^{\infty} a_n x^{n+1} + \sum_{n=0}^{\infty} 3^n x^{n+1} = 2xA(x) + a_0 + \frac{x}{1-3x},$$

$$A(x)(1-2x) = 2 + \frac{x}{1-3x} \implies A(x) = \frac{2-5x}{(1-2x)(1-3x)}.$$

Splitting into partial fractions, we have

$$A(x) = \frac{1}{1-2x} + \frac{1}{1-3x} = \sum_{n=0}^{\infty} 2^n x^n + \sum_{n=0}^{\infty} 3^n x^n,$$

so

$$a_n = 2^n + 3^n.$$

Problem 3

Setting $k = 2$ in the result from Problem 6, we have that

$$\sum_{n=0}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3}.$$

Substituting $x = \frac{1}{2}$, we get

$$\sum_{n=0}^{\infty} \frac{n^2}{2^n} = \frac{\frac{1}{2}(1+\frac{1}{2})}{(1-\frac{1}{2})^3} = \frac{(1/2)(3/2)}{(1/2)^3} = 6.$$

Problem 4

Let x be the decimal on the RHS, which is equal to

$$\sum_{n=1}^{\infty} \frac{F_{n-1}}{10^{2n}} = x.$$

By the Fibonacci recurrence,

$$x = \sum_{n=1}^{\infty} \frac{F_{n-1}}{10^{2n}} = \sum_{n=1}^{\infty} \frac{F_{n-2}}{10^{2n}} + \sum_{n=1}^{\infty} \frac{F_{n-3}}{10^{2n}} = \frac{1}{10^2} \sum_{n=1}^{\infty} \frac{F_{n-2}}{10^{2(n-1)}} + \frac{1}{10^4} \sum_{n=1}^{\infty} \frac{F_{n-3}}{10^{2(n-2)}},$$

$$\sum_{n=1}^{\infty} \frac{F_{n-2}}{10^{2(n-1)}} = \sum_{n=2}^{\infty} \frac{F_{n-2}}{10^{2(n-1)}} + \frac{F_{-1}}{10^0} = x + 1,$$

$$\sum_{n=1}^{\infty} \frac{F_{n-3}}{10^{2(n-2)}} = \sum_{n=2}^{\infty} \frac{F_{n-3}}{10^{2(n-2)}} + \frac{F_{-1}}{10^0} + \frac{F_{-2}}{10^{-2}} = x + 1 - 100 = x - 99,$$

$$x = \frac{1}{10^2}(x+1) + \frac{1}{10^4}(x-99),$$

$$10^4 x = 100(x+1) + (x-99) = 101x + 1,$$

$$(10^4 - 101)x = 1 \implies x = \frac{1}{9899}.$$

Problem 5

We have that

$$B(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} a_k \right) x^n = \sum_{n=0}^{\infty} \sum_{k=n}^{\infty} a_n x^k = \sum_{n=0}^{\infty} a_n \sum_{k=n}^{\infty} x^k$$

$$= \sum_{n=0}^{\infty} a_n \frac{x^n}{1-x} = \frac{A(x)}{1-x}.$$

Problem 6

$$\sum_{n=0}^{\infty} n^k x^n = \sum_{n=0}^{\infty} \left(x \frac{d}{dx} \right)^k (x^n) = \left(x \frac{d}{dx} \right)^k \sum_{n=0}^{\infty} x^n = \left(x \frac{d}{dx} \right)^k \left(\frac{1}{1-x} \right).$$

Let $k = 1$:

$$\sum_{n=0}^{\infty} n x^n = x \frac{d}{dx} \left(\frac{1}{1-x} \right) = \frac{x}{(1-x)^2}.$$

Let $k = 2$:

$$\sum_{n=0}^{\infty} n^2 x^n = \left(x \frac{d}{dx} \right) \left(\left(x \frac{d}{dx} \right) \left(\frac{1}{1-x} \right) \right) = x \frac{d}{dx} \left(\frac{x}{(1-x)^2} \right)$$

$$= -\frac{x^2 + x}{(x-1)^3}.$$

Problem 15

If $A(x) = \sum_{n=0}^{\infty} a_n x^n$, then $A(-x) = \sum_{n=0}^{\infty} (-1)^n a_n (-x)^n$, so

$$\frac{1}{2}(A(x) + A(-x)) = \sum_{n=0}^{\infty} a_{2n} x^{2n},$$

$$\sum_{n=0}^{\infty} a_{2n} x^n = \frac{1}{2}(A(\sqrt{x}) + A(-\sqrt{x})).$$

Also, we have that

$$\sum_{n=0}^{\infty} a_{n+1} x^n = \frac{1}{x} \sum_{n=0}^{\infty} a_{n+1} x^{n+1} = \frac{1}{x}(A(x) + A(0)),$$

so

$$\sum_{n=0}^{\infty} a_{2n+1} x^n = \frac{1}{2x}(A(\sqrt{x}) + A(-\sqrt{x}) - A(0)).$$

To find $\sum_{n=0}^{\infty} a_{3n} x^n$, let $\omega = e^{2\pi i/3}$. Then $1 + \omega + \omega^2 = 0$. Then

$$\sum_{n=0}^{\infty} a_{3n} x^{3k} = \frac{1}{3} \sum_{n=0}^{\infty} a_{3n}(1 + \omega^n + \omega^{2n}) x^{3n} = \frac{1}{3}(A(x) + A(\omega x) + A(\omega^2 x)).$$

So

$$\sum_{n=0}^{\infty} a_{3n} x^n = \frac{1}{3}(A(\sqrt[3]{x}) + A(\omega \sqrt[3]{x}) + A(\omega^2 \sqrt[3]{x})).$$